

★ TOPIC 00: ORIENTATION & PHILOSOPHY

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1 PURPOSE

Zero Fluff. Pure FAANG Signal.

Engineered specifically to get Senior & Staff SWE offers at Google, Meta, Amazon, Uber, Datadog & Airbnb.

Outcome:
Instant
pattern
recognition
+ zero-bug
Java
templates.

2 TARGET AUDIENCE

 **Beginners:** Intuitive mental models without heavy math proofs.

 **FAANG Candidates:** Master NeetCode 150 & LC Hard patterns.

 **Staff Engineers:** 20-min pre-interview template cram sheet.

3 WHY THIS HANDBOOK IS DIFFERENT

University / Textbooks	This FAANG Handbook
Isolated proofs & history	Pattern Evolution & DNA
Generic pseudocode	Production-Grade Java 17+
Memorizing 1000 problems	29 Core Pattern DNA Cards
Ignores interviews	Verbatim Interview Scripts

💡 AHA MOMENT

Pattern DNA > Problem Count:
Solving 1,000 random problems is a trap. Understanding 29 Pattern DNA cards solves thousands automatically!

🏆 GOLDEN RULE

CLARIFY → THINK → CHOOSE DS → OPTIMIZE → CODE → EXPLAIN. Never start coding without stating your pattern!

☕ JAVA ONLY

100% Java Collections API (ArrayDeque, PriorityQueue, HashMap). No generic pseudocode excuses.

⚡ SPEED DRILL

Write every template from memory in under 3 minutes. Zero syntax hesitation under pressure.

✦ **LEARNING PHILOSOPHY:** "Build intuition first, formalize into a template second, dry run on edge cases third." Turn the page to see the complete 20-Topic Dependency Graph!



TOPIC 00: SEQUENCED ROADMAP & DEPENDENCY TREE

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PHASE 1: LINEAR & SEARCH

00 Orientation & Strategy

Start



01 Foundations (Big-O, Math, Bits)

Base



02 Arrays & Prefix Sum

Core



03 Strings & KMP / Hash

Core



04 Hashing & Frequency

Core



05 Two Pointers

Pattern



06 Sliding Window

Pattern



07 Binary Search & Bounds

Search

PHASE 2: STRUCTURES & TREES

08 Linked Lists (Fast/Slow)

Struct



09 Stack & Monotonic Stack

Mono



10 Queue & Monotonic Deque

Mono



11 Heap / PriorityQueue

Top-K



12 Trees (DFS / BFS / BST)

Heavy



13 Trie (Prefix Trees)

String

PHASE 3: GRAPHS, DP & META

14 Graphs (DFS, BFS, Topo, UF)

Heavy



15 Recursion & Backtracking

Search



16 Dynamic Programming

Heavy



17 Greedy Algorithms

Opt



18 Staff Data Structures (LRU/LFU)

Staff



19 Interview Mastery & Meta

Final

 **DEPENDENCY RULE:** Never jump to Graphs (14) or DP (16) without mastering Trees (12) and Recursion (15). DP is simply Recursion + State Caching!



HOW EVERY CHAPTER WORKS (THE 13-BLOCK FLOW)

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1 RECOGNITION

Trigger keywords in problem statements mapping instantly to this pattern.

2 MENTAL MODEL

Intuitive visual/physical representation of how the algorithm operates.

3 VISUALIZATION

Step-by-step state diagrams, ASCII trees, and pointer movement illustrations.

4 CORE THEORY

Mathematical invariants and precise structural mechanics behind the pattern.

5 JAVA TEMPLATE

Production-grade Java code skeleton (cross-referenced to Appendix A).

6 LINE-BY-LINE

Detailed explanation of every variable, loop condition, and base case.

7 DRY RUN

Line-by-line variable state trace table on concrete representative inputs.

8 COMPLEXITY

Time & Space complexity analysis with mathematical justification.

9 EDGE CASES

Tricky boundary conditions (null, empty arrays, duplicates, single elements).

10 COMMON MISTAKES

Top 3 failure points candidate make in live FAANG coding interviews.

11 INTERVIEW QUESTIONS

Senior-level follow-up questions asked by Google, Meta & Uber interviewers.

12 LEETCODE SET

Curated progression of representative problems (Easy → Medium → Hard).

13 REVISION SUMMARY SHEET

1-Page condensed cheat card for 20-minute rapid pre-interview revision.

 **CONSISTENCY GUARANTEE:** Every single chapter in Book 1 & Book 2 follows this exact 13-block sequence so you never waste cognitive energy figuring out document formatting!



TOPIC 00: STUDY GUIDE & EXECUTION PLANS

30-DAY FAST TRACK

For Experienced Revisers

- **Days 1–7:** Topics 01–07 (Arrays, Two Pointers, Window, BS)
- **Days 8–18:** Topics 08–14 (Trees, Tries, Graphs)
- **Days 19–25:** Topics 15–17 (Recursion, DP, Greedy)
- **Days 26–30:** Template Speed Drills & Meta Scripting

60-DAY STANDARD

Recommended for FAANG

- **Days 1–15:** Linear Structures & Search
- **Days 16–35:** Non-Linear (Trees, Heaps, Graphs)
- **Days 36–50:** Complete DP & Staff Structures
- **Days 51–60:** Book 2 Problems & Mock Loops

90-DAY MASTERY

For Beginners / Staff L6

- **Month 1:** Foundations & Linear Patterns
- **Month 2:** Trees, Graphs & Backtracking
- **Month 3:** Advanced DP, Segment Trees & System Sync

INTERVIEW WEEK & 3-HOUR CRAM STRATEGY

Timeframe	Action Plan	Target Resource
7 Days Before	Review all 29 Pattern DNA cards in Book 2 Section 2	Book 2 Section 2
3 Days Before	Write all 24 Master Java Templates from memory	Book 1 Appendix A
1 Day Before	Read "If You See... Think..." Tables & Common Mistakes	Part 13 (Interview Meta)
3 Hours Before	Run 3-Hour Cram Sequence (Topic 00 Page 3)	Topic 00 Page 3 Table

✓ **READINESS CHECKLIST:** You have completed Topic 00! You understand the 20-Topic Dependency Graph, the 13-Block Chapter Architecture, and your personal Study Schedule. Turn the page to **Topic 01: Foundations (Big-O, Math & Bit Manipulation)**!